

RACHEL GUPTA

rachel.gupta@student.ufv.ca | Abbotsford, British Columbia, Canada | www.linkedin.com/in/rachel-gupta-rg2004 | <https://github.com/thevoidshell>

PROFESSIONAL SKILLS

Programming & Development: Java, Python, JavaScript, HTML/CSS, C/C++ (coursework), Assembly (learning)

Cybersecurity: Penetration Testing (VAPT), Network Reconnaissance, Subdomain Enumeration, Directory Brute-Forcing, OWASP Top 10, XSS, CSRF, Access Control Testing

Tools & Platforms: Nmap, Burp Suite, Sublist3r, Gobuster, DIRB, Wappalyzer, SecurityHeaders, WHOIS, Kali Linux, IDA Pro, Git, Docker

Infrastructure: Windows Server 2022 (AD, DHCP, DNS), Linux Command Line, Microsoft Office, Google Workspace

Soft Skills: Problem-Solving, Technical Communication, Team Collaboration, Detail-Oriented, Adaptable

PROFESSIONAL EXPERIENCE

Housing Desk Assistant

December 2025 – Present

UFV Housing Operations | Abbotsford, Canada

- Provided front-line customer service to 200+ students/guests, handling inquiries, promoting events via bulletin boards.
- Managed check-in/out, access cards/keys, CRM payments, mail sorting, phone coverage, and security logs.
- Monitored building access, enforced community standards, reported concerns, and escorted contractors.

Cybersecurity Intern

May 2025 – August 2025

RezoAI | Noida, India

- Developed advanced offensive security skills through an intensive, hands-on internship.
- Conducted vulnerability assessments and penetration testing (VAPT) on live web applications using Nmap, Burp Suite, and OWASP methodology
- Documented and communicated findings, provided actionable recommendations for risk mitigation.
- Collaborated with technical teams to enhance application and infrastructure security.

Research & Development and Production Specialist

February 2024 – July 2024

SortEZ | Chandigarh, India

- Led project teams focused on production process optimization and consistency.
- Pioneered initiatives that reduced raw material wastage from 15% to nearly zero within three months.
- Standardized workflow procedures, improving team learning and project outcomes.

PROJECTS

Explainable AI in Malware Detection

Python, Random Forest, SHAP, LIME, Machine Learning

- Developed a malware detection research project using a Random Forest classifier trained on Portable Executable (PE) static features derived from the EMBER dataset.
- Implemented SHAP and LIME explainability techniques to analyze model predictions and improve interpretability of malware classification decisions.
- Conducted a user study with 20 participants to evaluate how explainable AI impacts user confidence in cybersecurity decision-making.
- Analyzed entropy-based and structural file features to identify meaningful indicators of malicious behavior.

- Produced a formal research paper documenting methodology, statistical analysis, and interpretability findings.

Apache HTTP Server Installation, Configuration & Hardening

Kali Linux, Apache2, OpenSSL, Linux Administration

- Installed and configured Apache HTTP Server from source on Kali Linux, including virtual hosts, SSL/TLS encryption, logging, and firewall configuration.
- Implemented server hardening measures including restricted directory access, disabled directory indexing, minimized server token disclosure, and UFW firewall rules.
- Configured HTTPS using self-signed SSL certificates and validated encrypted communication through testing and monitoring.
- Developed backup and recovery procedures and implemented centralized logging and monitoring for operational visibility.
- Documented the full deployment, hardening, and recovery process in a technical systems administration report.

CERTIFICATIONS

Cybersecurity

- *PortSwigger Web Security Academy*: XSS (Jun 2025), CSRF (Jul 2025), Access Control (Jul 2025)
- *TryHackMe*: Pre-Security, Web Fundamentals (May 2025)
- *OverTheWire*: Bandit Linux Privilege Escalation (May 2025)

Infrastructure

- *LinkedIn Learning*: Windows Server 2022 Essential Training – Installation, AD, DHCP/DNS (Nov 2024)

Web Development

- *University of Michigan*: Advanced Responsive Design (Dec 2022), JavaScript Interactivity (Nov 2022), CSS3 (Nov 2022), HTML5 (Oct 2022)

EDUCATION

Bachelor of Computer Information Systems | CGPA: 3.90

University of the Fraser Valley | Abbotsford, British Columbia

Dean's List: Fall 2023 – Summer 2024 (Consecutively awarded)

CAMPUS INVOLVEMENT

Secretary – UFV Dance Club

January 2026 – Present

Social Media Manager – UFV Dance Club

January 2025 – December 2025

- Designed and executed social media strategies to boost event participation and engagement.

Volunteer Leader – Serving Humanity Club

September 2023 – July 2024

- Led donation and clean-up drives, benefiting 150+ individuals; organized and participated in CSR initiatives.

ACHIEVEMENTS & LEADERSHIP

First Prize Winner – UFV Community Ideation Jam

November 2024

- Proposed and developed an innovative community wellness solution as part of a multidisciplinary team.

Crowdfunding Leader – GreenSole Foundation

May 2022

- Raised INR 37,900 to supply 42 school kits to underprivileged children.

LEO Lion – Lions Club Chandigarh Central

November 2019 – November 2021

- Organized registration and outreach for medical camps, advancing local health efforts.